

STAT 9120: Theory of Linear Models

Take-Home Midterm Examination

Fall 2026 – 24-hour window

Name: _____ **Submitted:** _____

Take-home instructions. This examination is open-book and open-note. You may use the course text, lecture notes, posted homework, and standard computational software to *check arithmetic*. All submitted derivations, proofs, and statistical arguments must be shown clearly and must be your own work. Do not collaborate or discuss the examination with others during the 24-hour window. When a theorem or identity is central to your argument, identify it explicitly. Unsupported numerical answers may receive limited credit.

The 24-hour window is intended to provide scheduling flexibility; the examination is designed for substantially less than 24 hours of active work. Unless otherwise stated, assume the normal linear model

$$Y = X\beta + \varepsilon, \quad \varepsilon \sim N_n(0, \sigma^2 I).$$

Problem	1	2	3	4	5	6
Points	18	18	14	18	20	12

Total: 100 points

Problem 1. Rank deficiency, estimability, and parameterization 18 points

Consider a four-group effects model with two observations in each group,

$$Y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, 2, 3, 4, \quad j = 1, 2,$$

with parameter vector

$$\beta = (\mu, \alpha_1, \alpha_2, \alpha_3, \alpha_4)^T$$

and design matrix

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

- (a) Find a basis for $\mathcal{N}(X)$ and determine $\text{rank}(X)$. [3 points]
- (b) Let $c = (c_0, c_1, c_2, c_3, c_4)^T$. Derive a single linear condition on the entries of c that is necessary and sufficient for $c^T \beta$ to be estimable. Explain briefly why this condition follows from the null-space characterization of estimability. [4 points]
- (c) Determine whether each of the following is estimable. Justify each answer using the condition from part (b):

$$\mu, \quad \alpha_1 - \alpha_2, \quad \mu + \alpha_3, \quad 2\mu + \alpha_1 + \alpha_2, \quad \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4.$$

[5 points]

(d) Suppose the four observed group sample means are

$$\bar{Y}_1 = 3, \quad \bar{Y}_2 = 5, \quad \bar{Y}_3 = 4, \quad \bar{Y}_4 = 8.$$

Because the group sizes are equal, the least-squares fitted group means are these four sample means. Find one least-squares parameter vector under each identifying constraint:

$$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 0 \quad \text{and} \quad \alpha_4 = 0.$$

Verify directly that the two parameter vectors give the same fitted group means and the same value of the estimable contrast $\alpha_1 - \alpha_2$, despite assigning different values to individual parameters. Explain the statistical point illustrated by this calculation. [6 points]

Problem 2. Model updating and sequential covariate addition 18 points

A base model with full-rank design matrix X has residual vector

$$r = M_X Y = \begin{pmatrix} 2 \\ -1 \\ 1 \\ -2 \end{pmatrix}.$$

Two candidate covariates have already been residualized against the base model:

$$z^* = M_X z = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \quad w^* = M_X w = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \end{pmatrix}.$$

Thus all three vectors lie in $\mathcal{C}(X)^\perp$.

(a) Add z to the base model. Using the Frisch–Waugh–Lovell result, compute the coefficient of z and the reduction in SSE. [4 points]

(b) Now add w *after* z . First residualize w^* against z^* :

$$w_{|z} = w^* - z^* \frac{z^{*T} w^*}{z^{*T} z^*}.$$

Compute $w_{|z}$, the coefficient of the added w direction, and the additional reduction in SSE. [5 points]

(c) Reverse the order: add w first, and then residualize z^* against w^* . Compute the two successive reductions in SSE. Show that the *total* reduction in SSE is the same in either order, even though the two sequential reductions differ. Find the final SSE, starting from the base-model SSE $r^T r$. Explain geometrically why the final answer cannot depend on the order in which z and w are added. [5 points]

(d) In an unrelated fitted model, suppose

$$\hat{\beta} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad (X^T X)^{-1} x_i = \begin{pmatrix} 1/4 \\ -1/8 \end{pmatrix}, \quad e_i = \frac{8}{5}, \quad h_{ii} = \frac{3}{5}.$$

Using

$$\hat{\beta}_{(i)} = \hat{\beta} - \frac{(X^T X)^{-1} x_i e_i}{1 - h_{ii}}, \quad e_{(i)} = \frac{e_i}{1 - h_{ii}},$$

compute $\hat{\beta}_{(i)}$ and $e_{(i)}$. Explain briefly why leverage appears in both deletion formulas. [4 points]

Problem 3. Multivariate normal transformations and chi-squared distributions

14 points

Suppose

$$Y \sim N_4(\mu, \Sigma), \quad \mu = \begin{pmatrix} 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 3 \end{pmatrix}.$$

Define

$$U_1 = Y_1 + Y_2, \quad U_2 = Y_1 - Y_2, \quad U_3 = Y_3 - Y_4.$$

(a) Find the joint distribution of $U = (U_1, U_2, U_3)^T$. Give its mean vector and covariance matrix. **[5 points]**

(b) Are U_1, U_2, U_3 mutually independent? Justify your conclusion carefully, stating the property of the multivariate normal distribution you use. **[2 points]**

(c) Determine the distribution of

$$Q = \frac{(U_1 - 2)^2}{6} + \frac{U_2^2}{2} + \frac{(U_3 - 2)^2}{4}.$$

State the degrees of freedom and justify your answer.

[3 points]

(d) Determine the distribution of

$$Q_0 = \frac{U_1^2}{6} + \frac{U_2^2}{2} + \frac{U_3^2}{4}.$$

If the distribution is noncentral, give the noncentrality parameter. Explain the difference between Q and Q_0 . **[5 points]**

Problem 4. Nested projections, Cochran's theorem, and the partial F test

18 points

Let X_0 and X_1 be full-column-rank design matrices satisfying

$$\mathcal{C}(X_0) \subset \mathcal{C}(X_1),$$

with projection matrices H_0 and H_1 and ranks $p_0 < p_1$. Define

$$P = H_1 - H_0, \quad M_1 = I - H_1.$$

(a) Show that $H_1 H_0 = H_0$ and $H_0 H_1 = H_0$. Use these identities to prove that P is a symmetric idempotent matrix. Identify the subspace onto which P projects. **[5 points]**

(b) Show that

$$\text{rank}(P) = p_1 - p_0$$

and that

$$P M_1 = 0.$$

Interpret both facts geometrically.

[4 points]

(c) Assume

$$Y \sim N_n(\mu, \sigma^2 I)$$

and that the reduced-model null hypothesis is true, so $\mu \in \mathcal{C}(X_0)$. Using the projection quadratic-form result and Cochran's theorem (or an equivalent orthogonal-Gaussian argument), derive the distributions of

$$\frac{Y^T P Y}{\sigma^2} \quad \text{and} \quad \frac{Y^T M_1 Y}{\sigma^2},$$

and show that they are independent. Hence derive the null distribution of the usual partial F statistic. [5 points]

(d) Numerically, suppose

$$n = 40, \quad p_0 = 3, \quad p_1 = 6, \quad SSE_0 = 84, \quad SSE_1 = 60.$$

Compute the partial F statistic. Using $F_{3,34;0.95} = 2.88$, state the conclusion at level $\alpha = 0.05$. [4 points]

Problem 5. General linear hypotheses, nested models, and confidence regions 20 points

Consider a full-rank normal linear model with

$$n = 30, \quad p = 4, \quad MSE = 2,$$

$$\hat{\beta} = \begin{pmatrix} 2 \\ 3/2 \\ -1/2 \\ 1 \end{pmatrix}, \quad (X^T X)^{-1} = \begin{pmatrix} 1/5 & 0 & 0 & 0 \\ 0 & 1/4 & 1/8 & 0 \\ 0 & 1/8 & 1/4 & 0 \\ 0 & 0 & 0 & 1/2 \end{pmatrix}.$$

Let

$$C = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix}, \quad d = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus the null hypothesis is $H_0 : C\beta = d$.

(a) Interpret the two restrictions in words. What is $q = \text{rank}(C)$? [2 points]

(b) Compute $C\hat{\beta}$, $C(X^T X)^{-1}C^T$, the quadratic form

$$Q = (C\hat{\beta} - d)^T [C(X^T X)^{-1}C^T]^{-1} (C\hat{\beta} - d),$$

and the general linear F statistic. Using $F_{2,26;0.95} = 3.37$, state the decision at level $\alpha = 0.05$. [7 points]

(c) Since $SSE_1 = (n - p)MSE$, compute the full-model SSE. Treating Q as the extra sum of squares associated with these two restrictions, find the corresponding reduced-model SSE and verify that the nested-model partial F statistic is identical to the result in part (b). [4 points]

(d) Let

$$\theta = C\beta.$$

Write the 95% joint confidence region for θ in quadratic-form form. Determine whether $\theta = (0, 0)^T$ lies in the region, and explain why your answer must agree with the hypothesis-test conclusion in part (b). [4 points]

- (e) For the single restriction $H_0 : \beta_2 - \beta_3 = 0$, compute the usual t statistic. Verify numerically that the corresponding one-degree-of-freedom F statistic satisfies $F = t^2$. Use $t_{26;0.975} = 2.056$ if you wish to state a test decision. [3 points]

Problem 6. Structural statements: prove, disprove, or qualify 12 points

For each statement below, decide whether it is true as written. If it is true, give a concise proof. If it is false, give a counterexample or state the missing condition and explain why it matters. Each part is worth 3 points.

- (a) If A is symmetric and $Z \sim N_n(0, I)$, then $Z^T A Z$ has a chi-squared distribution.

- (b) If $P^2 = P$, then for $Z \sim N_n(0, I)$,

$$Z^T P Z \sim \chi_{\text{rank}(P)}^2.$$

No symmetry assumption is needed.

- (c) If random variables U and V satisfy $\text{Cov}(U, V) = 0$, then U and V are independent.

- (d) In a rank-deficient linear model, imposing an identifying constraint makes every linear function $c^T \beta$ testable.